Four nested limit cycles in a planar power-law reaction system

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Abstract

We prove that a planar power-law reaction system considered by Boros and Hofbauer admits a stable positive equilibrium surrounded by four nested hyperbolic limit cycles. From the innermost to the outermost, their stability alternates between repelling and attracting. The system therefore supports a stable resting state and two stable oscillations at the same parameter values. The proof starts from a two-parameter reduction of the Hopf and first-focus conditions. Directed integer interval arithmetic isolates a common zero of the next two focus coefficients, proves that the ninth-order radial coefficient is negative, and verifies a nonsingular four-parameter unfolding. A separate real Lyapunov-polynomial calculation with directed decimal intervals confirms the decisive root, sign, and rank conditions. A parameter-dependent finite normal form and a uniform continuously differentiable remainder estimate then produce four simple fixed points of the return map of the original analytic differential equation. All cycles lie in a compact annulus strictly inside the positive quadrant. The result is existential and local: it does not specify a finite-amplitude parameter example or a physical period, and it allows real reaction orders.

Keywords: limit cycles; power-law kinetics; weak focus; Hopf bifurcation; interval arithmetic.
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1 Introduction

Uniqueness of a positive steady state does not preclude several attracting oscillations in a nonlinear reaction model. A particularly economical example was proposed by Boros and Hofbauer [3, Section 5.1]. They identified approximate parameters at which three focus values vanish and a fourth is negative, and left the existence of four limit cycles as a conjecture. We establish the corresponding existence statement for their specified four-parameter family.

The issue is more than the accuracy of a numerical weak-focus location. One must prove that the simultaneous zero exists, certify the sign of the next coefficient, verify enough independent parameter directions, and control the difference between a finite normal form and the original equation. We supply these four ingredients. The finite coefficient calculations are checked in two different representations and two different arithmetic implementations; the final passage to periodic orbits uses the true return map.

Multiple nested reaction-system cycles, including alternating stability, are already known in other families. In particular, Craciun and Erban [4, Theorem 1, equations (4.27)–(4.30), and Section 5] construct nested algebraic cycles using polynomial products. Our contribution concerns the particular sparse power-law family below, rather than the general coexistence pattern or the general method of unfolding a weak focus. The calculation does not depend on the optional notebook cited by Boros and Hofbauer, whose contents were unavailable for comparison. No exhaustive priority claim about that material is made.

2 Model and result

Consider, in the original dimensionless time t,

$$\dot{x} = y^{-1} - x^a y^b, \qquad \dot{y} = K\left[-(d-1) - y^{-1} + dx^a y^b\right] \tag{2.1}$$

on $\mathbb{R}^2_{>0}$, with parameter domain

$$\mathcal{P} = \{(K, a, b, d) : K > 0,\ a > 0,\ b > -1,\ d > 1,\ 1 + bd > 0\}. \tag{2.2}$$

These are the normalized equations and inequalities of [3, Section 5.1], with $d > 1$ enforcing existence of the positive steady state. Real powers have their usual positive-real meaning. Consequently the vector field is jointly real analytic in state and parameters on its stated domain. No boundary condition on an axis is imposed or needed for the local construction.

Solving the first equilibrium equation gives $x^a y^b = y^{-1}$; the second then gives $(d-1)(y^{-1} - 1) = 0$. Thus the unique positive equilibrium is $(1, 1)$. Its Jacobian, trace, and determinant are

$$J = \begin{pmatrix} -a & -(1+b) \\ Kda & K(1+bd) \end{pmatrix}, \qquad T = -a + K(1+bd), \qquad \Delta = Ka(d-1) > 0. \tag{2.3}$$

Theorem 1. *There exist parameters in $\mathcal{P}$ for which the unique positive equilibrium of* (2.1) *is hyperbolically asymptotically stable and is surrounded by at least four disjoint nested hyperbolic periodic orbits. From the innermost to the outermost, the four orbits are repelling, attracting, repelling, and attracting. They are all contained in a compact annulus in $\mathbb{R}^2_{>0}$ that has positive distance from $(1, 1)$ and from the coordinate axes.*

The parameters will approach a rigorously isolated fourth-order weak focus p_*, approximately

$$\begin{aligned} a_* &\simeq 1.0128219986803146, & b_* &\simeq 0.6546256211042382, \\ d_* &\simeq 3.2886104393099358, & K_* &\simeq 0.3212443603989608. \end{aligned} \tag{2.4}$$

These rounded values locate the organizing weak focus, not a parameter example with four cycles. The cycles occur at nearby parameters constructed in Section 6. Hyperbolicity refers to the nontrivial multiplier of each periodic orbit.

3 Focus coefficients and reduction

Write $q = 1 + bd$. On the Hopf hypersurface $K = a/q$, put $\omega^2 = \Delta$ and set

$$u = x - 1, \quad v = y - 1, \quad X = u, \quad Y = \frac{au + (1+b)v}{\omega}, \quad z = X + \mathrm{i}Y. \tag{3.1}$$

The linear field is $\dot{z} = \mathrm{i}\omega z$ with positive angular orientation. All coefficient definitions below retain time t.

For $i + j \geq 1$, the coefficients of the original real Taylor jet are

$$\begin{aligned} f_{ij} &= \mathbf{1}_{i=0}\binom{-1}{j} - \binom{a}{i}\binom{b}{j}, \\ g_{ij} &= K\left[-\mathbf{1}_{i=0}\binom{-1}{j} + d\binom{a}{i}\binom{b}{j}\right]. \end{aligned} \tag{3.2}$$

Here $\binom{\alpha}{0} = 1$ and $\binom{\alpha}{j+1} = \binom{\alpha}{j}(\alpha - j)/(j+1)$; the constant coefficients cancel. Setting $s = \mathrm{i}\omega$, substitute

$$u = \frac{z + \bar{z}}{2}, \qquad v = \frac{(-a-s)z + (-a+s)\bar{z}}{2(1+b)}, \qquad \mathcal{F} = f + \frac{s}{\omega^2}\left[af + (1+b)g\right]. \tag{3.3}$$

Computations in the algebra $s^2 = -\omega^2$ use rational operations only.

We fix the following finite normal-form convention. Find $H = z + O(|z|^2)$ and $G = sz + O(|z|^2)$ satisfying, through degree nine,

$$\mathcal{F}(H, \bar{H}) = H_z G + H_{\bar{z}} \bar{G}. \tag{3.4}$$

All resonant coefficients $H_{m+1,m}$ are set to zero. If degrees below n have been determined, let $\mathcal{R}_n$ be the degree-n residual of the left side minus the right side of (3.4). For $j + k = n$, the next coefficients are

$$\begin{cases} G_{jk} = (\mathcal{R}_n)_{jk}, \quad H_{jk} = 0, & j - k = 1, \\ H_{jk} = \dfrac{(\mathcal{R}_n)_{jk}}{(j-k-1)s}, \quad G_{jk} = 0, & j - k \neq 1. \end{cases} \tag{3.5}$$

Indeed, adding H_{jk} changes this residual by $-(j-k-1)sH_{jk}$, and adding G_{jk} changes it by $-G_{jk}$. Every nonresonant denominator is nonzero. Define

$$F_m = \operatorname{Re} G_{m+1,m}, \qquad 1 \leq m \leq 4. \tag{3.6}$$

If $F_1, \ldots, F_{m-1}$ vanish, the first nonzero radial return coefficient is $(2\pi/\omega)F_m$. This follows by writing $z = re^{\mathrm{i}\theta}$ and integrating the first radial drift over one positive angular revolution. Coefficients beyond the first nonzero one need not agree across different normal-form conventions.

The degree-three instance of (3.2)–(3.5) gives the rational identity

$$F_1 = \frac{a(2a^2 d + aq - q^2)}{16q^2}. \tag{3.7}$$

After multiplying by $2\pi/\omega$, this is the first focus value printed in [3, Section 5.1]. Introduce $h > 2$ by

$$d = \frac{h(h-1)}{2}, \qquad b = \frac{ah - 1}{d}, \qquad K = \frac{1}{h}. \tag{3.8}$$

Then $q = ah$, so $T = F_1 = 0$ identically. Taking the derivative with respect to b on the Hopf hypersurface first, and then imposing (3.8), yields

$$\partial_b F_1|_{T=0,\,(3.8)} = -\frac{(h-1)(2h-1)}{32h} < 0. \tag{3.9}$$

Thus eliminating the first focus equation has not removed its transverse parameter direction. In what follows $f(a, h) = (F_2, F_3)$ denotes the restriction to (3.8).

3.1 Exact calibrations

The calculation has two useful published checks. For the system of [3, equation (12), Proposition 3],

$$\begin{aligned} \dot{x} &= y^{-q_c} - xy^{1/2}, \\ \dot{y} &= K_c[-(q_c + r_c + 1/2) + (q_c + 1/2)y^{-q_c} + r_c xy^{1/2}], \end{aligned} \tag{3.10}$$

at $(q_c, r_c, K_c) = (1/4, 15/8, 4/3)$, the convention above gives

$$\omega^2 = \frac{7}{8}, \qquad F_1 = F_2 = 0, \qquad F_3 = -\frac{4375}{884736}, \qquad \frac{2\pi F_3}{\omega} = -\frac{625\pi}{110592}\sqrt{\frac{7}{2}}. \tag{3.11}$$

For the endpoint $(a, h) = (1, 22/7)$ of the target family, corresponding to $d = 165/49$, exact rational arithmetic instead gives

$$F_1 = F_2 = 0, \qquad F_3 = -\frac{18125}{2404233216} < 0. \tag{3.12}$$

The zero and sign in (3.12) agree with the boundary statement in [3, Section 5.1]; its displayed fraction is our calculation in convention (3.6). The two benchmark systems are distinct.

4 An interval certificate for the weak focus

The following two finite decimals are exact rational inputs:

$$\begin{aligned} a_0 &= 1.012821998680314600068208268899051820974352006, \\ h_0 &= 3.112895114354931694231478861591838762886663535. \end{aligned} \tag{4.1}$$

Let $x_0 = (a_0, h_0)$ and $\delta = 10^{-20}$. For the interval calculation we use an outward dyadic enclosure $\mathcal{X}$ of $x_0 + [-\delta, \delta]^2$, with grid spacing 2^{-384}. The exact rational preconditioner is given entrywise by

$$\begin{aligned} R_{11} &= -523.05136623435780257862812824527858797 09, \\ R_{12} &= 1724.43841383885651080539417225644379593 5, \\ R_{21} &= 1632.40120139576008685936756199968616876 3, \\ R_{22} &= -3959.57512664182068880590301715606501281. \end{aligned} \tag{4.2}$$

Its determinant is nonzero by exact rational arithmetic. Neither the center nor the preconditioner is assumed accurate in advance.

Proposition 2. *The box $x_0 + [-10^{-20}, 10^{-20}]^2$ contains a unique common zero (a_*, h_*) of (F_2, F_3). At this zero $F_4 < 0$, and $\det \mathrm{D}_{(a,h)}(F_2, F_3) \neq 0$. The corresponding parameters* (3.8) *belong to $\mathcal{P}$.*

Proof. Evaluate (3.2)–(3.5) with integer interval endpoints on the 2^{-384} grid. Addition and subtraction are exact on this grid. For multiplication and division, all endpoint products or quotients are considered and rounded outward using integer floor and ceiling. A division is allowed only when the denominator interval excludes zero. All divisions in this calculation satisfy that check. First derivatives with respect to a, h are propagated by the product and quotient rules along the same recurrence. The resulting interval matrix $J_f(\mathcal{X})$ therefore encloses $\mathrm{D}f(x)$ at every $x \in \mathcal{X}$.

Table 1 records convenient rational outer bounds on the resulting intervals. Full integer endpoints and the executable recurrence are supplied in Online Resource 1. In particular, the table is a rounded outward display of strict enclosures, not floating-point error estimates.

Table 1: Directed dyadic enclosures on $\mathcal{X}$. Every displayed interval has been widened from the exact computed endpoints.

Quantity	Rational outer bound
$(J_f)_{11}$	$[0.005322623692367449,\ 0.005322623692367584]$
$(J_f)_{12}$	$[0.002318061020176081,\ 0.002318061020176173]$
$(J_f)_{21}$	$[0.002194340814884875,\ 0.002194340814887489]$
$(J_f)_{22}$	$[0.000703107152964103,\ 0.000703107152965913]$
$\det J_f$	$[-1.344241117333,\ -1.344241117316]\,10^{-6}$
F_4	$[-3.54349145,\ -3.54349141]\,10^{-7}$
First row sum of $\lvert I - RJ_f\rvert$	$< 3.872\,10^{-12}$
Second row sum of $\lvert I - RJ_f\rvert$	$< 8.938\,10^{-12}$
Both coordinates of $\lvert\mathcal{Y} - x_0\rvert$	$< 9\,10^{-32}$

Here the interval image is

$$\mathcal{Y} = x_0 - Rf(x_0) + (I - RJ_f(\mathcal{X}))(\mathcal{X} - x_0). \tag{4.3}$$

The map $\Phi(x) = x - Rf(x)$ maps $\mathcal{X}$ into $\mathcal{Y}$ by the integral mean-value formula. The row-sum bounds make it a strict contraction in the maximum norm, and the image bound places $\mathcal{Y}$

strictly inside the original rational box. Banach's fixed-point theorem gives one and only one fixed point in $\mathcal{X}$. Since R is nonsingular, this is exactly a zero of f. The fixed point belongs to $\mathcal{Y}$, so it also lies in the original box without dyadic enlargement.

The determinant and F_4 intervals prove the two remaining coefficient claims. Direct interval evaluation of (3.8) gives $a > 0$, $b > -1$, $d > 1$, $K > 0$, $1 + bd > 0$, and $\omega^2 > 0$ on the whole box. For example, deliberately coarse enclosures are $1 < a < 1.02$, $0.65 < b < 0.66$, $3.28 < d < 3.30$, $0.32 < K < 0.33$, and $0.74 < \omega^2 < 0.75$. This also establishes admissibility. □

The code uses only the Python standard library for strict certification. Its input is (4.1)–(4.2), not previously calculated focus values. A separate real scalar recurrence and directed decimal interval implementation certify the same zero and signs; Appendix A states its conventions and bounds.

5 A full parameter unfolding

First extend the finite normal form smoothly off the Hopf hypersurface. For $p = (K, a, b, d)$ near p_*, set

$$\tau = T/2, \qquad \nu = \sqrt{\Delta - \tau^2} > 0, \qquad X = u, \quad Y = \frac{(a+\tau)u + (1+b)v}{\nu}. \tag{5.1}$$

The linear field is $(\tau X - \nu Y, \nu X + \tau Y)$. With $\lambda = \tau + \mathrm{i}\nu$, use $j\lambda + k\bar{\lambda} - \lambda$ as the nonresonant denominator in (3.5), still retaining all terms with $j - k = 1$. For $j - k \neq 1$, the imaginary part $(j - k - 1)\nu$ is bounded away from zero after restricting the parameter neighborhood. This is a finite polynomial change of variables depending smoothly on p. Its linear part is the identity, so it is a local diffeomorphism on a uniform sufficiently small state neighborhood.

The actual transformed field, including its remainder, has the form

$$\dot{z} = (\tau + \mathrm{i}\nu)z + \sum_{m=1}^{4} C_m(p) z|z|^{2m} + E_p(z), \tag{5.2}$$

where $\operatorname{Re} C_m = F_m$ on $T = 0$. On a sufficiently small closed parameter neighborhood and state ball there is a uniform constant C such that

$$|E_p(z)| \le C|z|^{10}, \qquad \|\mathrm{D}_z E_p(z)\| \le C|z|^9. \tag{5.3}$$

These estimates follow from Taylor's theorem for the jointly analytic original field and the finite coordinate change on a compact set. They do not assume convergence of an infinite normal-form series.

Define

$$\beta(p) = (\beta_0, \beta_1, \beta_2, \beta_3) = (T/2, \operatorname{Re} C_1, \operatorname{Re} C_2, \operatorname{Re} C_3). \tag{5.4}$$

We next verify that these four coefficients are local parameter coordinates. Replace (K, a, b, d) by (T, b, a, h), where $K = (T + a)/(1 + bd)$ and $d = h(h-1)/2$. On $T = 0$, the first-focus zero is the graph in (3.8). By the chain rule, or equivalently the Schur complement,

$$\det \frac{\partial(F_1, F_2, F_3)}{\partial(b, a, h)} = (\partial_b F_1) \det \mathrm{D}_{(a,h)} f \tag{5.5}$$

at p_*. Accounting for $\tau = T/2$ and the parameter order gives

$$\det \mathrm{D}_{(K,a,b,d)} \beta(p_*) = -\frac{1 + bd}{2h - 1} (\partial_b F_1) \det \mathrm{D} f(a_*, h_*) \neq 0. \tag{5.6}$$

The factors are nonzero by (3.9) and Proposition 2; a strict outer enclosure of the determinant is $[-8.990, -8.989]\,10^{-8}$. Derivatives of C_m in the T direction do not change this triangular determinant argument. The inverse function theorem therefore applies to β at p_*.

The unfolding varies a, b, d as well as K: reaction powers and a reaction-vector component change. It is not an unfolding by rate constants of a single fixed network. Thus it does not contradict the rate-constant codimension bound in [1, Theorem 35]. Likewise, the inheritance theorem in [2, Theorem 3.2] assumes an already transverse source bifurcation; it does not replace the coefficient and rank calculation above.

6 Four cycles of the original analytic equation

Proof of Theorem 1. Let $\ell = F_4(p_*) < 0$ and $\omega_* = \sqrt{\Delta(p_*)} > 0$. By (5.6), for every sufficiently small $\varepsilon > 0$ there is a fixed parameter vector $p(\varepsilon)$ satisfying

$$\begin{aligned} \beta_0 &= 24\ell\varepsilon^8, & \beta_1 &= -50\ell\varepsilon^6, \\ \beta_2 &= 35\ell\varepsilon^4, & \beta_3 &= -10\ell\varepsilon^2. \end{aligned} \tag{6.1}$$

Moreover $p(\varepsilon) - p_* = O(\varepsilon^2)$, so $\operatorname{Re} C_4(p(\varepsilon)) = \ell + O(\varepsilon^2)$. These parameters remain in the open set $\mathcal{P}$. The equilibrium has eigenvalues $\beta_0 \pm \mathrm{i}\nu$ with $\beta_0 < 0$, hence is hyperbolically stable.

Put $z = \varepsilon R e^{\mathrm{i}\theta}$ and consider a fixed compact R interval away from zero, such as $[1/2, 5/2]$, together with a slightly larger interval for controlling the flow. Equations (5.2)–(6.1) imply $\dot\theta = \omega_* + O(\varepsilon^2) > 0$. Define

$$P(R) = R(R^2-1)(R^2-2)(R^2-3)(R^2-4). \tag{6.2}$$

The identity $\prod_{j=1}^{4}(S-j) = S^4 - 10S^3 + 35S^2 - 50S + 24$ and the uniform remainder bounds give

$$\frac{dR}{d\theta} = \varepsilon^8 \frac{\ell}{\omega_*} P(R) + O_{C^1}(\varepsilon^9). \tag{6.3}$$

The notation O_{C^1} means a uniform bound on the function and its first R derivative, also uniform in θ. In particular, the radial projection of E_p is $O(\varepsilon^{10})$; dividing by ε makes its contribution $O(\varepsilon^9)$, with the same order for the R derivative by (5.3). The angular-speed correction and the change in C_4 contribute only $O(\varepsilon^{10})$ to (6.3).

Let Π_ε be the actual return map on $\theta = 0$, after an increase of 2π in θ. Its definition uses solutions of (5.2) including E_p, hence solutions of (2.1). Over this fixed angular interval, R changes by $O(\varepsilon^8)$. Integrating (6.3) and its variational equation gives

$$\frac{\Pi_\varepsilon(R) - R}{\varepsilon^8} = \frac{2\pi\ell}{\omega_*} P(R) + O_{C^1}(\varepsilon). \tag{6.4}$$

For completeness, a uniform bound on the right side of (6.3) keeps the flow in the slightly larger interval through 2π; its R derivative is $O(\varepsilon^8)$, so the variational solution is $1 + O(\varepsilon^8)$ by Gronwall's inequality. Substitution into the integral equations yields both estimates in (6.4). No long-time numerical approximation is involved.

The four positive roots $R_j = \sqrt{j}$ of P are simple. Choose four disjoint small closed intervals around them. On each interval the limiting displacement in (6.4) has opposite endpoint signs and a derivative of constant nonzero sign. Uniform C^1 convergence preserves these properties for sufficiently small fixed $\varepsilon > 0$. There is exactly one simple return-map fixed point in each of the four intervals.

The derivatives $P'(R_j)$ are $-12, 8, -12, 48$ in order. Since $\ell < 0$, the signs of $\Pi'_\varepsilon - 1$ at the four fixed points are $+, -, +, -$. Each multiplier is positive and approaches one as $\varepsilon \to 0$, so these signs give the asserted alternating repelling and attracting stability. Positive

angular speed makes each periodic orbit wind once around the equilibrium. Uniqueness of planar solutions makes the four Jordan curves disjoint and nested. The local coordinate transformations preserve this geometry and orbital stability.

Finally fix one sufficiently small positive ε. All four curves lie in a neighborhood of $(1,1)$ with closure in the positive quadrant. Their compact union avoids the equilibrium and the axes. Choosing an inner Jordan curve inside the innermost orbit and an outer Jordan curve outside the outermost orbit gives the required compact annulus. This proves the theorem for the original equation. □

6.1 Physical meaning and limitations

At the constructed parameters, a stable rest state and two attracting oscillations coexist in the same dimensionless two-concentration model. The two repelling cycles are local separators of the nearby attracting dynamics. The result concerns a neighborhood of the steady state; it gives neither global permanence nor an upper bound on the total number of cycles in the positive quadrant.

The finite normal form determines a sufficiently small parameter unfolding, while analyticity supplies the existence of a uniform remainder constant. We have not evaluated that constant to give a numerical upper bound for ε. Thus (2.4) must not be used as a directly simulated four-cycle example. Computing explicit finite-amplitude parameters and enclosing their four return roots is a further task. No physical period is reported. The argument retains the dimensionless time in (2.1); a positive monomial change of time would preserve orbits but would not preserve periods. Finally, real reaction orders and the permitted reaction-vector variation do not establish an integer-stoichiometric chemical implementation or an experimentally observed oscillator.

A Independent scalar Lyapunov calculation

A second calculation avoids the complex conjugacy recurrence. On the Hopf hypersurface use $X = u$, $Z = au + (1+b)v$, $w = \omega^2$, and $Q = wX^2 + Z^2$. The linear field is $(-Z, wX)$. Reconstruct the real field directly from (3.2) and $v = (Z - aX)/(1+b)$ and seek

$$H = Q + \sum_{n=3}^{10} H_n, \qquad \nabla H \cdot (\dot{X}, \dot{Z}) = \sum_{m=2}^{5} c_m Q^m + O(\|(X,Z)\|^{11}), \tag{A.1}$$

where H_n is homogeneous of degree n. Although H is computed through degree ten, only the degree-nine vector-field jet is needed.

Let B_n be the contribution at degree n of the previously computed H_k and the nonlinear field, and let $L = -Z\partial_X + wX\partial_Z$. For odd n solve $LH_n = -B_n$. For $n = 2m$, set c_m equal to the angular mean of B_n on $Q = 1$ and solve $LH_n = c_m Q^m - B_n$. The required averages are

$$\left\langle X^{2p} Z^{2q} \right\rangle_{Q=1} = \frac{(2p)!(2q)!}{4^{p+q} p! q! (p+q)! w^p}, \tag{A.2}$$

and are zero when either exponent is odd. If $H_n = \sum_{j=0}^{n} h_j X^{n-j} Z^j$, the scalar recurrence is

$$w(j+1)h_{j+1} - (n-j+1)h_{j-1} = (\mathrm{rhs})_j, \tag{A.3}$$

with coefficients outside $0 \le j \le n$ equal to zero. At even degree impose $h_0 = 0$ to fix the radial kernel; at odd degree the last equation fixes h_0. These equations give an independent finite coefficient generator.

The scalar and complex coefficients have a triangular relation. If $F_1 = \cdots = F_{k-1} = 0$, the first nonzero term satisfies

$$c_{k+1} = \frac{2F_k}{w^k}. \tag{A.4}$$

To see this, pull H into the finite complex normal form. Its quadratic term is wr^2, whose first radial derivative term is $2wF_k r^{2k+2}$. Division by $Q^{k+1} = w^{k+1}r^{2k+2}$ gives (A.4). Changes in lower radial kernel terms add only multiples of lower focus values. This proves a smooth invertible lower-triangular change between the two focus sequences. In particular, on their common lower-zero locus,

$$\det \mathrm{D}(c_3, c_4) = 4w^{-5} \det \mathrm{D}(F_2, F_3), \qquad F_4 = \tfrac{1}{2}w^4 c_5. \tag{A.5}$$

The derivative determinants here use the reduced variables (a, h). These equalities are used at the simultaneous zero; a general off-zero high-order coefficient is not identified across gauges.

The second implementation uses 90-digit decimal endpoints, with floor and ceiling rounding at every interval operation. It computes its own exact rational preconditioner by inverting the rational midpoint of its center derivative enclosure. Its only shared numerical proposals are (4.1) and the radius 10^{-20}. Table 2 summarizes its strict outer bounds.

Table 2: Independent directed-decimal calculation. The last two rows use (A.5) only at the certified common zero.

Quantity	Rational outer bound
Two contraction row sums	$< 2.802\,10^{-11}$, $< 6.423\,10^{-11}$
Two image displacements	$< 2.802\,10^{-31}$, $< 6.423\,10^{-31}$
$\det \mathrm{D}(c_3, c_4)$	$[-2.3487381622, -2.3487381619]\,10^{-5}$
c_5	$[-2.305150908, -2.305150598]\,10^{-6}$
F_4 at the common zero	$[-3.543491671, -3.543491195]\,10^{-7}$
$\det \mathrm{D}(F_2, F_3)$ at that zero	$[-1.344241118, -1.344241117]\,10^{-6}$

The same contraction argument establishes the zero independently. Exact rational evaluations also give $c_2 = c_3 = 0$ and $c_4 = -625/42336$ for (3.10), reproducing (3.11), and reproduce (3.12). All homogeneous-equation residuals in those rational checks vanish exactly. The independent implementation does not import the complex coefficient generator, its arithmetic module, its preconditioner, or cached focus outputs.

Statements and declarations

Data and code availability. Online Resource 1 contains the two coefficient generators, directed arithmetic implementations, exact rational certificate inputs, reference output bounds, and instructions for reproducing the certificates and exact calibrations. No experimental data are used. No third-party paper is included in the supplementary package.

Use of artificial intelligence. OpenAI Codex was used substantively for literature analysis, candidate derivations, implementation of the calculations, computational checks, and drafting. A separate AI session implemented the scalar verification described in Appendix A. These automated checks are not external peer review and do not establish completed human scientific review. Human review of the reasoning, code, citations, and final text remains pending for this local draft.

Author declarations. Final author confirmation of affiliation, contributions, funding, competing interests, and approval of the manuscript is pending.

References

[1] Murad Banaji, Balázs Boros, and Josef Hofbauer. Bifurcations in planar, quadratic mass-action networks with few reactions and low molecularity. *Nonlinear Dynamics*, 112: 21425–21448, 2024. https://doi.org/10.1007/s11071-024-10068-1.

[2] Murad Banaji, Balázs Boros, and Josef Hofbauer. The inheritance of local bifurcations in mass action networks. *Journal of Nonlinear Science*, 35:72, 2025. https://doi.org/10.1007/s00332-025-10165-4.

[3] Balázs Boros and Josef Hofbauer. Oscillations in planar deficiency-one mass-action systems. *Journal of Dynamics and Differential Equations*, 36(Suppl 1):175–197, 2024. https://doi.org/10.1007/s10884-021-10051-z. First published online in 2021.

[4] Gheorghe Craciun and Radek Erban. Planar chemical reaction systems with algebraic and non-algebraic limit cycles. *Journal of Mathematical Biology*, 90:64, 2025. https://doi.org/10.1007/s00285-025-02221-0.